



**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE AND TECHNOLOGY**

**DAQA GUIDELINES FOR CURRICULUM
DEVELOPMENT AND REVIEW**



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1.0 INTRODUCTION

The World has continued to experience change. These changes can broadly be classified as Political, Economic and Social. Have these changes been necessary? This question drives one to the cause(s) of change. Ever since, man has strived to live better. Being interdependent and dependent on the environment, there has always been a need to understand self, one another and the environment. The vast world, depletion of non- renewable resources accompanied with greed has created a need to create fast means of traversing the globe and communicate, search for alternative means or efficient and effective utilization of resources and creation of institutions to curb and beat the vice. This introduces the subject Education.

The definition of Education has been evolving. Bloom's taxonomy seems widely embraced. I agree that Education is the process of acquiring knowledge, skills and attitudes. These captures the cognitive, psychomotor and affective domains. It is not surprising that Kenya has experienced a drastic change in the nature and structure of the education system since independence. The Ominde commission (1964), the Gachathi commission (1976) and the Mackay commission are among the commissions set to identify and incorporate changes in the Kenyan form of Education in order to make it relevant to change.

1.1 Objectives of University Education (*Universities act, no.42, 2012, sec 3*)

(1) The objectives of university education shall include:

- (a) Advancement of knowledge through teaching, scholarly research and scientific investigation;
- (b) Promotion of learning in the student body and society generally;
- (c) Promotion of cultural and social life of society;
- (d) Support and contribution to the realization of national economic and social development;
- (e) Promotion of the highest standards in, and quality of, teaching and research;
- (f) Education, training and retraining higher level professional, technical and management personnel;
- (g) Dissemination of the outcomes of the research conducted by the university to the general community;
- (h) Facilitation of life-long learning through provision of adult and continuing education;
- (i) Fostering of a capacity for independent critical thinking among its students;
- (j) Promotion of gender balance and equality of opportunity among students and employees; and
- (k) Promotion of equalization for persons with disabilities, minorities and other marginalized groups.



- (2) In the discharge of its functions and the exercise of its powers under this Act, a university shall be guided by the national values and principles of governance set out under Article 10 of the Constitution, and shall in that regard:
- (a) Promote quality and relevance of its programmes;
 - (b) Enhance equity and accessibility of its services;
 - (c) Promote inclusive, efficient, effective and transparent governance systems and practices and maintenance of public trust;
 - (d) Ensure sustainability and adoption of best practices in management and institutionalization of systems of checks and balances;
 - (e) Promote private-public partnership in university education and development; and
 - (f) Institutionalize non-discriminatory practices.

1.2 Summary of the Objectives

The objectives the University Education as spelled in the Universities act of 2012 can be summarized as falling into four major categories:

- i. Cognitive development of the Learners
- ii. Affective development of the Learners
- iii. Skills development of the Learners
- iv. Fostering quality governance in Universities.

2.0 THE EDUCATIONAL CURRICULUM

Schedule III of the CUE standards and guidelines, 2014, provides standards intended for use by universities in Kenya in the development, implementation, quality assurance and review of academic programmes. Here, an “*academic programme*” means the design of learning content, which is multi-dimensional and includes intentions, structure of content, delivery modes, academic resources and assessment modes. The dimensions implied here are not clarified. A “*programme of study*” means the prescribed syllabus that students must be taught at each key stage. This is synonymous with the dictionary definition of the word curriculum.

The above definition leads to a limited concept of the curriculum, defined in terms of what teaching and instruction is to be offered, purpose, resources and examination i.e assessment. This kind of definition fails to take account of the educational or moral dimensions of the educational curriculum (A.V. Kelly, The Curriculum, Theory and Practice, sixth Edition, 2010, page 7).

2.1 The Total Curriculum

Learning can take place in formal or informal settings and any experience that has a formative effect on the way one thinks, feels, or acts may be considered educational. ‘ We must recognize



the distinction often drawn between the ‘ formal’ curriculum and the ‘informal’ curriculum, between the formal activities for which the timetable of the school allocates specific periods of teaching time and those many informal activities that go on , usually on a voluntary basis, at lunch- times, after school hours, at weekends or during holidays. These latter activities often called ‘extracurricular’ activities are suggested to be seen as separate from, as over and above the curriculum itself’. (A.V. Kelly, The Curriculum, Theory and Practice, sixth Edition, 2010, page 12). These activities can be identified with guidance and counselling, sports and games, educational and social clubs in Universities. They have been found to contain much educational validity and point as any formal arrangements of the University. This implies a proper definition of ‘educational curriculum’ must include the totality of the experiences the student has as a result of the provision made.

2.2 The Centrality of the Lecturer

The Lecturers have a core responsibility in curriculum Planning, development, implementation evaluation and innovation.

3.0 CURRICULUM DEVELOPMENT

The following are very critical to consider for any educational curriculum:

- i. Curriculum planning
- ii. Curriculum development
- iii. Curriculum design
- iv. Curriculum implementation

3.1 Tyler’s Rationale

In 1949, R . W. Tyler produced his course entitled ‘Basic Principles of Curriculum and Instruction’. In it he identified four fundamental questions which, he suggested, should be answered in developing any curriculum and plan of instruction. These questions are:

- i. What educational purposes should the educational establishment seek to attain?
- ii. What educational experiences can be provided that are likely to attain these purposes?
- iii. How can these educational experiences be effectively organized?
- iv. How can we determine whether these purposes are being attained?

The Tyler rationale continues to be the basis of the recent curriculum development which has taken place. Attempts have been made to extend the list of possible questions, comparing the curriculum process (for example) with a journey, as is shown below:

1. Is the journey necessary?



2. Where are we going (Aims and objectives?)
3. What road do we take? (Subject model)
4. What vehicle shall we drive? (Content)
5. How shall we drive the vehicle? (Approach to learning)
6. What sort of map shall we provide? (Educational technology)
7. Who are our fellow travelers? (Other areas of the curriculum)
8. How do we tell whether we are on the right track? (Evaluation)
9. How can we tell if we have arrived? (Assessment)
10. How do we tell others? (Dissemination)
11. What mistakes did we make? (Feedback)

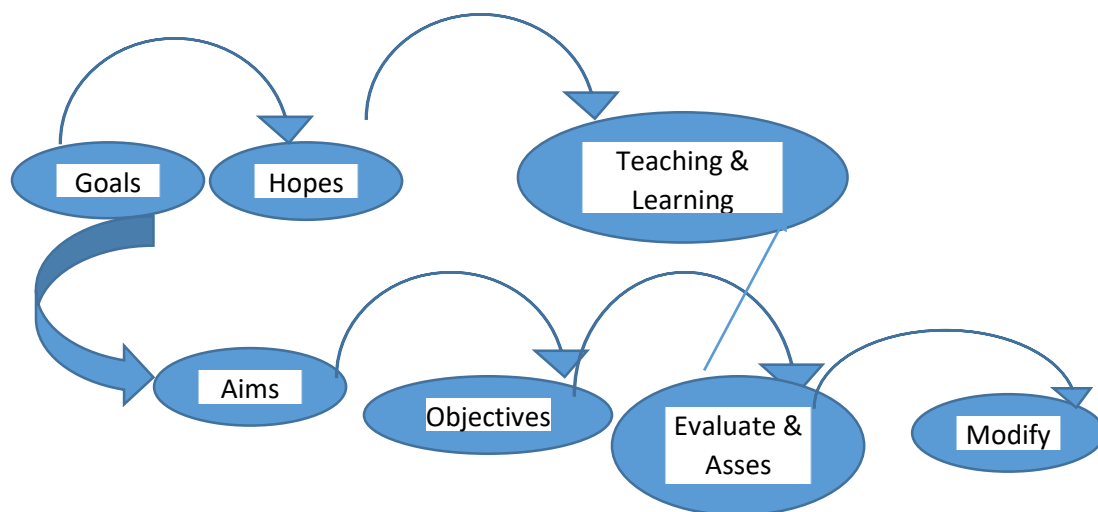


Fig. 3.1 A model for the curriculum process

Figure 3.1 assumes that "goals" are the general outcomes of a course of study. Some of these are translated into long term hopes ("... to produce a professional doctor") and others into more detailed aims which are applicable to the whole of a course. These are the aims which are assessed (e.g. by examination) . By assessment one means 'how' successfully have students achieved the course aims? "; Evaluation covers the efficacy of teaching, feedback from students, and evaluation of the assessment procedures.



3.1.1 The Aims of a Course

When aims are being selected, the curriculum developer must take account of a number of "pressures", including:

- i. The development (social, psychological, physical and emotional) of the student
- ii. The nature of the subject being taught
- iii. The requirements of the consumer (e.g. Business and industry)
- iv. Restraints (e.g. Cultural and economic) imposed by resources
- v. The competence of the teaching staff.

If the pressures are ignored, a course can be produced which is both difficult to effect and is irrelevant to the student.

3.1.2 Objectives

For day to day teaching purposes, aims must be turned into specific teaching objectives. These objectives are not a list of things to learn, they are not a syllabus (in the accepted sense of the term) and they say nothing about the teaching approach. Objectives show "what a student should be able to do as a result of a learning experience.

Mager gives the following characteristics of a "behavioral" objective: (Mager R.F, 1961)

1. They specify the kind of behavior which will be accepted as evidence that the learner has achieved the objective.
2. They describe the important conditions under which the behavior will be expected to occur.
3. They describe how well the learner must perform to have his behavior considered acceptable.

In phrasing objectives, certain terms (e.g. to know, to understand, to appreciate, to be concerned) are open to many interpretations and so are probably best not used. More suitable terms, which are open to fewer interpretations, include: to write, to recall, to solve, to list.

4.0 BLOOM'S TAXONOMY OF LEARNING DOMAINS

Bloom's Taxonomy was created in 1956 under the leadership of educational psychologist Dr Benjamin Bloom in order to promote higher forms of thinking in education, such as analyzing and evaluating concepts, processes, procedures and principles rather than just remembering facts. It is used when designing educational,



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training and learning processes. It can be used as the basis for choice of curriculum objectives.

4.1 Bloom's Learning Domains

Bloom's committee identified three *domains* of educational activities or learning (Bloom, et al. 1956):

1. **Cognitive:** Mental skills (knowledge)
2. **Affective:** Growth in feelings or emotional areas (attitude or self)
3. **Psychomotor** : Manual or physical skills (skills)

This classification of learning behaviors may be thought of as “the goals of the learning process.” That is, after a learning episode, the learner should have acquired a new skill, knowledge, and/or attitude.

4.1.1 The Cognitive Domain

The cognitive domain involves knowledge and the development of intellectual skills. This includes the recall or recognition of specific facts, procedural patterns, and concepts that serve in the development of intellectual abilities and skills. There are six major levels of cognitive domain processes, starting from the simplest to the most complex:

- i. Knowledge
- ii. Comprehension
- iii. Application
- iv. Analysis
- v. Synthesis
- vi. Evaluation

The levels can be thought of as building on one another. That is, the first ones must normally be mastered before the next one can take place.

4.1.2 Bloom's Cognitive Revised Taxonomy

Lorin Anderson, a former student of Bloom, and David Krathwohl revisited the cognitive domain in the mid-nineties and changed the names in the six categories from noun to verb forms.



The revised Bloom's Revised Taxonomy has the following hierarchical categories:

1. Creativity
2. Evaluation
3. Analysis
4. Application
5. Understanding
6. Remembering

The curriculum objectives must be specific, measurable, attainable, realistic and time bound (SMART). They must also be hierarchical from first year to the final year.

5.0 COMPONENTS OF THE CURRICULUM

There are four components in a curriculum:

1. The purpose (goals and objectives)
2. Content or subject matter
3. Methods or learning experiences
4. Evaluation

5.1 Purpose

The curriculum consists of aims, goals and objectives. The objectives are usually stated in terms of expected outcomes. There are three categories of goals and objectives:

- i. Cognitive- knowledge, comprehension, application, analysis, synthesis and evaluation.
- ii. Affective- receiving, responding, valuing, organization and characterization.
- iii. Psychomotor- perception, set, guided response, mechanism, complex overt response, adaptation and origination.

5.2 Content or Subject matter

This includes the knowledge to be learned. This is divided into bodies of knowledge, desired attitudes and values and skills.

- a. The criteria used in selection of subject matter for the curriculum should be:
 - i. Self-sufficiency- this includes more results and effective learning outcomes in the most economical manner.



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- ii. Significance- contribute to basic ideas to achieve overall aim of curriculum, develop learning skills.
- iii. Validity- meaningful to the learner based on maturity, prior experience, educational and social value.
- iv. Usefulness- usefulness of the content either for the present or the future.
- v. Learnability- within the range of the experience of the learners.
- vi. Feasibility- can be learned within the time allowed, resources available, expertise of the teacher and nature of the learner.

b. Principles to follow in organizing the learning contents:

- i. **Balance**- the content should be fairly distributed in depth and breadth of the particular learning or discipline. This ensures that the level or area of learning is not overloaded or inadequate.
- ii. **Articulation**- each level of subject matter should be smoothly connected to the next, glaring gaps or wasteful overlaps in the subject matter will be avoided.
- iii. **Sequence**- this the logical arrangement of the subject matter. It refers to the deepening and broadening of content as it is taken up in the higher levels.

5.3 Methods or Learning experiences

Teaching strategies convert the written curriculum to instruction. Whatever methods used to implement the curriculum, there will be some guide for the selection and use such as :

- i. Teaching methods are a means to achieve the end.
- ii. Teaching methods should stimulate the learner's desire to develop the cognitive, affective, psychomotor, social and spiritual domain of the individual.
- iii. In the choice of teaching methods, learning styles of the learners should be considered.
- iv. Flexibility should be considered in the choice of teaching methods.

5.4 Evaluation

To be effective, every curriculum must have an element of evaluation which is used to:

- i. Select appropriate content based on the aims and objectives of the curriculum.
- ii. Select appropriate teaching methods to deliver the content in order to achieve programme purpose.
- iii. Check the effectiveness of methods and learning experiences used.



- iv. Check on the suitability and the appropriateness of the curriculum in answering social needs.
- v. Give feedback to the planners, learners, teachers, industry, and society.
- vi. Provide a rationale for making change.

6.0 BASIS FOR CURRICULUM DEVELOPMENT / REVIEW

Any curriculum developed must have a clearly defined purpose. The purpose will crystalize by:

- i. A scientific needs assessment
- ii. Identification of stakeholders and an effective collection of their views.
- iii. Data collection and analysis using approved statistical methods and packages.

6.1 CUE guidelines and Standards on Curriculum Development

Schedule III of the CUE standards and guidelines (2014) , outlines standards and guidelines on University academic programme development. Some of these are:

i. PROG/STD/01

An academic programme shall facilitate a balanced learning process, ensuring that the students are able to acquire such cognitive, affective and psychomotor skills as are consistent with educational goals and aspirations of Kenyans.

ii. PROG/STD/02

Each level of academic programme shall be differentiated by specific attributes. Higher levels of academic programmes shall require more complex attributes according to Bloom's Taxonomy.

iii. PROG/STD/04

An academic programme structure shall be aligned to a standard format as provided for by the Commission and shall include background information of the university, information on the various programme facets and details on the academic resources for the support of the programme.

6.2 Data collection on Curriculum development and Review

For a curriculum to be relevant, effective collection of data and analysis must be done. This will be aided by development of appropriate information collection tools and appropriate analysis of the data collected. The form of information to be collected will be guided by but not limited to:

- i. National economic blueprints.
- ii. Prudent identification of stakeholders and diligent collection of their views.



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- iii. A diligent needs assessment.

6.3 Development of appropriate questionnaires

Questionnaires are proven information collection tools. Appropriate questionnaires for collection of information on curriculum development or review must address the following areas:

- i. The curriculum purpose
- ii. The curriculum content
- iii. The Teaching and Learning methods
- iv. The facilities and methods to aid the learning experience
- v. Learning assessment methods
- vi. Evaluation of the total curriculum.

6.4 Model of Questionnaires

The department must develop suitable questionnaires for collection of appropriate information either for curriculum development or review. The framing of the questions will be distinctive i.e. whether it is a review or development. The following are some of the questions that can serve as guides to the form of questions to be answered.

6.4.1 Curriculum development

A curriculum should be developed upon need assessment and collection of the stakeholders' views.

6.4.1.1 Curriculum purpose

- i. Is the curriculum addressing the national economic blueprints?
- ii. Is there a need identified that the curriculum will fulfill?
- iii. Has there been consultation with stakeholders who have indicated need for a particular curriculum?
- iv. Is the identified need short or long term? This will determine whether to develop a short course / diploma/ or degree programme.
- v. What will be required to make the graduates have an international orientation?
Remember the curriculum must develop graduates for consumption i.e tailored for an identified existing market.



6.4.1.2 Curriculum content

The curriculum content will be determined by the curriculum goals, objectives and expected learning outcomes.

- i. What are the objectives of the course? e.g. to produce a public health officer/ a medical doctor (practitioner)/ a nurse/ a medical researcher/ a pharmacist.
- ii. Is the course to be developed regulated by a registered regulator? What does the regulator require?
- iii. What will be the required cognitive, affective and psychomotor skills?
- iv. How will the content be organized to be hierarchical in line with the Bloom's taxonomy?
- v. What are the CUE standards and regulations for such a course? eg the number of contact hours?

6.4.1.3 Teaching methods and Learning experiences

This should be based on each individual courses, the curriculum objectives and should be learner centered.

- i. Which teaching methods will suit the content delivery? The teaching methods should motivate the learners with a cybernetic approach.
- ii. What facilities and methods will be used to give the learner a classroom experience that will mimic the work experience?
- iii. Which teaching methods will enhance learning?
- iv. Which facilities will facilitate learning?
- v. Which methods will enhance affective and psychomotor skills?

6.4.1.4 Assessment methods

Assessment methods should be based on the course objectives and learner centered.

- i. Which methods will effectively assess the cognitive, affective and psychomotor skills?
- ii. Which assessment methods will be used to determine whether the course objectives are attained?
- iii. Were the assessment methods adequate in grading the learners? This can be answered by the lectures partially but majorly by identifying an external assessment system to subject your students to for purposes of comparison.



6.4.2 Curriculum Review

Curriculum review should be carried out upon normative and summative evaluation. The outcomes of the curriculum purpose, curriculum objectives and the expected learning outcomes will determine the type of review to be carried out.

6.4.2.1 Curriculum purpose

Develop questionnaires for stakeholders that will investigate whether the curriculum purpose was achieved. Statistics on the following areas will give ideas on achievement of the curriculum purpose:

- i. The level of employability of the graduates in both public and private sectors.
- ii. The number of self-employed students in curriculum related areas. These is the number of business enterprises set by the graduates (Human capital) after identifying a need in the society in line with the curriculum purpose.
- iii. The progressive high school student programme preference as indicated by (UCCPB) Universities and colleges central placement board. This should be monitored during a full programme cycle.
- iv. The student transition rate as indicated by student progression from first year through the final year.

6.4.2.2 Curriculum content

The departmental quality assurance committee should develop questionnaires for students/alumni/lecturers for collection of statistics on:

- i. The hierarchical progression of the courses.
- ii. The relevance of the courses to the curriculum purpose.
- iii. The adequacy of the content for the set programme period. This should be checked against the CUE standards and guidelines.
- iv. The comprehensiveness of the content to meet the cognitive, affective and psychomotor domains education as pertaining the curriculum.
- v. The achievement of the programme objectives through the course objectives.

6.4.2.3 Teaching methods and Learning experiences

The departmental quality assurance committee should develop questionnaire to collect information on:

- i. Whether the teaching methods employed were adequate to enable the learner get the expected knowledge, develop the anticipated attitudes and acquire the necessary skills.



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- ii. Whether the teaching methods and facilities employed enabled the learner to get a learning experience mimicking the work experience.
- iii. Whether there was adequate communication from the lecturer to impact a positive learning attitude on the learner.
- iv. Whether there were adequate tutorials / assignments/ practicals/workshops to enhance a learning experience for the learner.
- v. Whether there was classroom learner participation to enable student-lecturer interaction.
- vi. Whether the teaching methods employed aroused a sense of analysis, application and creativity in the student.
- vii. Whether the teaching methods used integrated ICT.
- viii. Whether the teaching methods used elicited a sense of community service in the learner.

6.4.2.4 Assessment methods

Assessment methods used should grade a learner in relation with the expected skills, attitudes and knowledge. The questionnaires developed should provide information whether the assessment methods employed were adequate to determine:

- i. The level of cognitive skills of the Learner as per the Bloom's taxonomy.
- ii. The level of psychomotor skills of the Learner as per the Bloom's taxonomy
Vis a Vis the work environment.
- iii. The development of attitude in relation to work and social expectations of the learner.

6.4.2.5 Curriculum Evaluation

Curriculum evaluation is done on the basis of the course purpose with the curriculum content, the teaching methods and the student assessment methods serving as the drivers. The statistics to be collected will be based on the following key questions:

- i. Was the course purpose achieved? This can be evaluated by checking on the employability of the graduates and the private business enterprises started by the graduates which are both indicators of the business opportunities available.
- ii. Was the curriculum content adequate and relevant to the curriculum purpose? This can be answered by both the stakeholders and the graduates.
- iii. Were the teaching methods and learning experiences adequate to achieve the curriculum purpose? This is a question to be answered by the graduates.



- iv. Were the assessment methods adequate in grading the learners? This can answered by the lectures partially but majorly by identifying an external assessment system to subject your students to for purposes of comparison.
- v. What were the student transition and completion rates?
- vi. What was the programme selection preference by expected candidates compared to declared capacity over the years?

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